Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains one correct missing expected value e.g. 0 and car $= \frac{28 \times 35}{80}$	AO1.1a	M1	(see table below)
	Obtains all expected values Must not be rounded CAO Condone 12.937 for 12.9375	AO1.1b	A1	

	0	1	2	3+
C	12.25	10.0625	8.3125	4.375
MB	15.75	12.9375	10.6875	5.625

	Marking Instructions	AO	Marks	Typical Solution
(b)	States both hypotheses using correct language Accept equivalent wording	AO2.5	B1	H_0: There is no association between number of claims and type of insurance policy H_1: There is an association between number of claims and type of insurance policy
	Combines expected and observed values for 2 and 3 or more correctly Follow through their answers to part (a)	AO1.1b	B1F	(see table below)
	Calculates χ^2-test statistic Can be awarded even if 2 and 3 or more are not combined	AO1.1a	M1	$\sum \frac{(O-E)^2}{E} =$ $\frac{(9-12.25)^2}{12.25} + \frac{(19-15.75)^2}{15.75} + \frac{(10-10.0625)^2}{10.0625} + \frac{(13-12.9375)^2}{12.9375} + \frac{(16-12.6875)^2}{12.6875} + \frac{(13-163125)^2}{16.3125}$
	Calculates χ^2-test statistic correctly Condone given to 1 s.f. CAO	AO1.1b	A1	$= 3.07$
	States critical value (or p-value, follow through their χ^2 value) Condone χ^2 cv for 3 df = 6.251 if 2 and 3 or more are not combined	AO1.1b	B1F	χ^2 cv for 2 df = 4.605 ($p = 0.215$) 3.07 < 4.605 (0.215 > 0.1) Accept H_0
	Evaluates χ^2-test statistic by comparing the cv with their ts (or their p value with 0.1)	AO3.5a	R1	No significant evidence to suggest that there is an association between number of claims and type of insurance policy
	Infers H_0 not rejected (OE)	AO2.2b	E1	

	2 or more	
	Observed	Expected
C	16	12.6875
MB	13	16.3125

Concludes in context (not definite) Accept equivalent wording	AO3.2a	E1	
			Total 10 marks

Q2.

<table>
<tr><th></th><th>Marking Instructions</th><th>AO</th><th>Marks</th><th>Typical Solution</th></tr>
<tr><td rowspan="2">(a)</td><td>Obtains one missing expected value eg
Low and Breed A = $\frac{21 \times 27}{105}$</td><td>AO1.1a</td><td>M1</td><td rowspan="2">
<table>
<tr><th></th><th>Low</th><th>Med</th><th>High</th></tr>
<tr><td>A</td><td>5.4</td><td>9.6</td><td>6</td></tr>
<tr><td>B</td><td>5.14</td><td>9.14</td><td>5.71</td></tr>
<tr><td>C</td><td>8.23</td><td>14.63</td><td>9.14</td></tr>
<tr><td>D</td><td>8.23</td><td>14.63</td><td>9.14</td></tr>
</table>
</td></tr>
<tr><td>Obtains all expected values.
A0 if integers</td><td>AO1.1b</td><td>A1</td></tr>
<tr><td rowspan="5">(b)
(i)</td><td>States both hypotheses using correct language</td><td>AO2.5</td><td>B1</td><td rowspan="5">H_0 Milk yield is independent of breed
H_1 Milk yield is not independent of breed
1 tail 1%
$ts = \sum \frac{(O-E)^2}{E} = 19.4$
χ^2 cv for 6 df = 16.81
(p = 0.00275)
219.4 > 16.81
(0.00275 < 0.01)
Reject H_0 and conclude that there is evidence to suggest that milk yield is not independent of breed</td></tr>
<tr><td>States critical value (or p-value)</td><td>AO1.1b</td><td>B1</td></tr>
<tr><td>Evaluates the χ^2 test statistic by comparing the cv with the ts (or p-value with 0.01)</td><td>AO3.5a</td><td>R1</td></tr>
<tr><td>Infers H_0 rejected</td><td>AO2.2b</td><td>E1</td></tr>
<tr><td>Concludes correctly in context (conclusion should not be definite)</td><td>AO3.2a</td><td>E1</td></tr>
<tr><td rowspan="2">(ii)</td><td>Considers $\frac{(O-E)^2}{E}$ or (O – E) to identify largest sources of association
[Do not allow mark if no reference to why source selected]</td><td>AO2.4</td><td>E1</td><td>Largest sources of association
Breed A/High and Breed B/ Low.</td></tr>
<tr><td>Interprets main source(s) of association in context</td><td>AO3.2a</td><td>E1dep</td><td>Far more than expected Breed A cows observed to have high milk yield
or
Far more than expected</td></tr>
</table>

			Breed B cows observed to have low milk yield
			Total 9 marks

Q3.

<table>
<tr><th>Marking Instructions</th><th>AO</th><th>Marks</th><th>Typical Solution</th></tr>
<tr><td>States both hypotheses using correct notation</td><td>AO2.5</td><td>B1</td><td>H_0: Language choice is independent of sex</td></tr>
<tr><td>Constructs 'correct' contingency table with frequencies shown (PI)</td><td>AO3.1b</td><td>M1</td><td>H_1: Language choice is not independent of sex
<table>
<tr><th></th><th>German</th><th>Mandarin</th><th></th></tr>
<tr><td>M</td><td>7</td><td>12</td><td>19</td></tr>
<tr><td>F</td><td>23</td><td>8</td><td>31</td></tr>
<tr><td></td><td>30</td><td>20</td><td>50</td></tr>
</table></td></tr>
<tr><td>Finds expected frequencies (PI)</td><td>AO3.4</td><td>B1</td><td>
<table>
<tr><th></th><th>German</th><th>Mandarin</th></tr>
<tr><td>M</td><td>11.4</td><td>7.6</td></tr>
<tr><td>F</td><td>18.6</td><td>12.4</td></tr>
</table></td></tr>
<tr><td>Evaluates
$$\chi^2 = \sum \frac{(O-E)^2}{E}$$
Allow no Yates correction used

FT their observed and expected frequencies</td><td>AO1.1a</td><td>M1F</td><td>$$\chi^2 = \sum \frac{(|O-E|-0.5)^2}{E}$$
$$= \frac{(3.9)^2}{11.4} + \frac{(3.9)^2}{7.6} + \frac{(3.9)^2}{18.6} + \frac{(3.9)^2}{12.4}$$</td></tr>
<tr><td>Obtains χ^2 test statistic or quotes p-value</td><td>AO1.1b</td><td>A1</td><td>= 1.33 + 2.00 + 0.82 + 1.23

ts χ^2 = 5.38 p = 0.0204 1 tail 5%</td></tr>
<tr><td>Evaluates χ^2 model by comparing test statistic and correct critical value

Or by comparing p-value with 0.05</td><td>AO3.5a</td><td>M1</td><td>df = 1 cv = 3.84 ts > 3.84 or $p < 0.05$</td></tr>
<tr><td>Infers that H_0 should be rejected</td><td>AO2.2b</td><td>E1</td><td>Reject H_0</td></tr>
<tr><td>Completes test correctly and gives conclusion in context.

(conclusion should not be definite)</td><td>AO3.2a</td><td>E1</td><td>Significant evidence to suggest that language choice is not independent of sex</td></tr>
<tr><td colspan="4" align="right">Total 8 marks</td></tr>
</table>

EXAM PAPERS PRACTICE

Q4.

(a)

	C	L	LD	OP	Tot
M	156	144	120	60	480
F	216	135	108	81	540
Tot	372	279	228	141	1020

B1

For each correct row

B1

2

(b) H_0 : No association between gender and the way students vote

H_1 : Association between gender and the way students vote

For at least H_0 correct

B1

O_i	E_i	$(O_i - E_i)^2 / E_i$
156	175.06	2.075
216	196.94	1.844
144	131.29	1.230
135	147.71	1.093
120	107.29	1.505
108	120.71	1.337
60	66.35	0.608
81	74.65	0.541
		$X^2 = 10.233$

Attempt at E_i

M1

Attempt at $(O_i - E_i)^2 / E_i$

M1

Attempt at $\sum\left(\frac{(O_i - E_i)^2}{E_i}\right)$

M1

awfw 10.2 to 10.3
(A1 dependent on all 3 method marks)

A1

$v = 3 \Rightarrow \chi^2\text{crit} = 11.345$

B1

ft on their v value

B1 ft

$X^2 < \chi^2$crit $\therefore$ accept H_0

A1

Accept claim at 1% level.
Evidence to suggest that the way students vote is independent of gender.

E1

9

[11]

Q5.

(a) H_0: Choice independent of gender

gender not associated with choice

B1

	Squash	Badminton	Archery	Hockey	
Male	5/3.5	16/14	30/24.5	19/28	
Female	4/5.5	20/22	33/38.5	53/44	M1

Combine Squash and Badminton

$E_i < 5$ (Similar categories)

M1

	S & B	Archery	Hockey	
Male	21/17.5	30/24.5	19/28	M1
Female	24/27.5	33/38.5	53/44	M1

χ^2 values

	S & B	Archery	Hockey	
Male	0.7000	1.2347	2.8928	M1
Female	0.4455	0.7857	1.8409	

χ^2calc = 7.90

(7.8 to 7.9)

A1

$v = 2$

B1

$\chi^2_{5\%}(2) = 5.991$

(on their v)

B1ft

10

Reject H_o

Sufficient evidence, at the 5% level of significance, to support an association between the choice of sport and gender

reject H_o and H_o stated

or

statement in context

(b) **More females and fewer males** chose

B1

to participate in hockey **than expected**

B1

2

[12]

Q6.

(a)(b)

	Won	Lost	
< 2 hours	28 22.63	15 20.37	43
≥ 2 hours	22 27.37	30 24.63	52
	50	45	95

Method for table

M1

Labelled correctly, allow >2

A1

Numerically correct

A1

Method for E's – their table

M1

3

H_0 No association between length of match and chance of Boris winning

H_1 Association between length of match and chance of Boris winning

Null hypothesis – may be implied by

clearly stated conclusion–generous, allow 1 –sided etc

B1

$$\sum \frac{(|O-E|-0.5)^2}{E} = 4.04$$

Attempt at $\Sigma(O-E)^2/E$

M1

Attempt at Yates' correction

m1

Correct application of Yates' Correction

m1

4.04 (4 ~ 4.10)

A1

c.v. χ_1^2 is 3.841

1 df

B1ft

3.841 or 3.84, their df

B1ft

Reject H_0, evidence of association between result and length of game.

ft their figures – needs all M Marks and must be compared with upper tail of χ^2 (maximum 8 out of 9 if method of constructing table is incorrect) – Needs clearly stated correct conclusion or hypothesis

A1ft

9

(c) There is evidence of association between result and length of game but contrary to Boris's belief he appears to be less likely to win longer games.

evidence of association

E1

Boris less likely to win long games (2 marks for this point if evidence of association clearly stated in (b)) Needs proportion probably implied for both marks

E1

2

[14]

Q7.

(a)

	0	1	2 ≥ 3	Total
<2	1478 1508.94	212 191.24	$\underbrace{21 \quad 6}_{27}$ $\underbrace{14.36 \quad 2.45}_{16.81}$	1717
>2	2830 2799.06	334 374.76	$\underbrace{21 \quad 1}_{21}$ $\underbrace{26.64 \quad 4.55}_{31.19}$	3185
Total	4308	546	41 7	4902

method for 4 E's

M1

Method for all E's

M1

Attempt to combine classes

M1

Correct method for combining classes - needs all previous M marks

m1

H_0 : No association between number of claims and time for which policy held.
H_1 : Number of claims associated with time for which policy held.

Correct H_0 - may be implied by correctly stated conclusion.

B1

$$\sum\frac{(O-E)^2}{E} = \frac{30.94^2}{1508.94} + \frac{30.94^2}{2799.06} +$$

Attempt at $\sum\frac{(O-E)^2}{E}$ *- allow spurious continuity correction etc*

M1

$$\frac{20.76^2}{191.24} + \frac{20.76^2}{354.76} + \frac{10.19^2}{16.81} + \frac{10.19^2}{31.19}$$

Correct use of $\sum\frac{(O-E)^2}{E}$ *- requires first two and 4th M mark*

m1

= 14.0

14.0 (13.9 to 14.1)

A1

c.v. χ_2^2 for 1% risk is 9.21

2 df

B1f.t.

9.21, disallow ± 9.21

B1f.t.

Reject H_0 . Conclude number of claims associated with time policy held.

Conclusion - must be compared with upper tail of χ^2.

A1f.t.

11

(b) (i), (ii) **A**

Observed members not frequencies

E1

	Claim	No claim
Male	412	2824
Female	181	1485

cao

M1

A1

B

Classes overlap

E1

	Claim	No claim
<25	280	676
≥ 25	314	3632

cao

M1

A1

C

Classes not complete

E1

9

	No claim	≤ £2000	>£2000
<25	676	135	145
≥ 25	3632	122	192

cao

M1

A1

[20]

Examiner reports

Q1.

In part (a), the vast majority of students scored full marks. Some students made a numerical error with one of the entries in the table but it was very rare to make more than one error. Some students truncated the value of 12.9375 to 12,937 as some calculators give the table with only five digits displayed for each number. Highlighting the numbers on the calculator and reading the full output on the calculator would avoid this error.

The majority of students scored at least four marks for part (b). The most common hypotheses stated by students were ‘no association’ for H_0 and ‘association’ for H_1. Some students used ‘independent’ and ‘not independent’ instead, or a mixture. Some students failed to mention the variables in the hypotheses. Many students failed to merge the columns for 2 and 3+ even though the expected value of 4.375, which is less than 5, was given. Some students made numerical slips when calculating the test statistic. Many compared their test statistic with a critical value rather than calculating a corresponding probability. A minority of students drew a sketch to compare the test statistic with the critical value. Some sketches were not sufficiently clear and lost a mark.

Q4.

As in previous series, this proved to be a good source of marks for many candidates. In part (a), it was disappointing to see the many candidates who did not seem to understand the term ‘contingency table’. Consequently, the 2 marks available were awarded to those candidates who filled in the table with the expected frequencies or retrospectively for those who calculated the expected frequencies correctly in part (b). Others, who simply copied the values from the table given in the question or who produced values which did not tally with the given totals, gained no credit.

In part (b), many excellent solutions were seen. However, in a few cases, the hypotheses were not stated and z or t values were stated instead of the required χ^2 value of 11.345. Also, rounding errors arising from the calculation of the expected frequencies often resulted in the calculated value of X^2 falling outside the acceptable range.

Q5.

This question produced many disappointing responses. In part (a), most candidates stated correct hypotheses and usually worked out the required values for the E_i. However, there were still some candidates who insisted on rounding these values to the nearest whole number; which is penalised. The calculations of the E_i. were usually followed by an incorrect assumption that ν = 3 and consequently the wrong χ^2 value was quoted from the tables. Although many candidates failed to realise that when E_i. < 5, similar categories must be combined, the majority went on to use the correct method for evaluating χ^2. The conclusions drawn were usually correct and often in context. However, some candidates still think, incorrectly, that the rejection of the null hypothesis proves that the alternative hypothesis must be true. Part (b) proved to be beyond all but the very best candidates. The vast majority failed to interpret their results found in part (a). Most candidates simply referred to the original data, usually stating that 'more females than males played hockey' rather than making the required reference to the expected and observed values.

Q6.

As usual this type of question was extremely well answered with the great majority of candidates constructing the contingency table correctly and most applying Yates' correction accurately. The most common error with Yates' correction was to use (O – E – 0.5) instead of (|O − E |−0.5). There were some problems with the null hypothesis. Apart from some nonsensical hypotheses such as 'no difference between the length of the game and the probability of winning' there were also some one-sided hypotheses such as 'the longer a match lasts the better the chance of Boris winning'. The hypothesis test can only test for 'association' although further interpretation is possible by examining the table.

Q7.

Candidates performed very well on this question. Part (b) was testing, but most candidates were able to make a reasonable attempt.